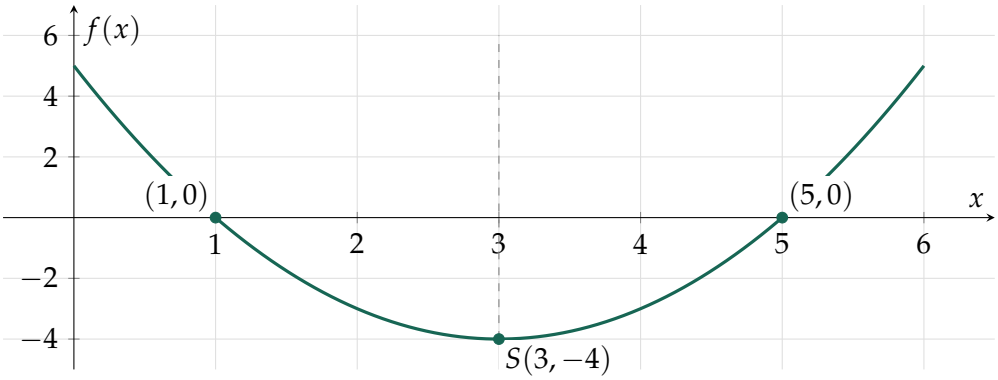
2.1 Three forms, three readings

Key point · Definition and results. A quadratic function is written $f(x) = ax^2 + bx + c$ with $a \neq 0$. The expanded form makes it easy to calculate $f(0)$; the vertex form $a(x-\alpha)^2 + \beta$ reveals the vertex and the extremum; a factorised form $a(x-r_1)(x-r_2)$ reveals the roots and the sign. The roots are the solutions of $f(x) = 0$.

Status and justification. Completing the square relies on $x^2 + 2ux = (x+u)^2 - u^2$. The existence of the vertex form is proved by expansion; its general computation is not an expected routine.

Example 1. $f(x) = x^2 - 6x + 5 = (x-3)^2 - 4 = (x-1)(x-5)$. The minimum is -4 at 3; the roots are 1 and 5; the y-intercept is 5.



Graph of $f(x) = (x-3)^2 - 4$. The two roots and the vertex are marked; the axis of symmetry is the line $x = 3$.

Watch out. A factorised expression is not obtained by putting each term separately in brackets.

Application. Give three forms of $g(x) = x^2 + 2x - 3$. Read off its roots and its minimum.

2.2 Discriminant and solving

Key point · Definition and results. For $ax^2 + bx + c = 0$ with $a \neq 0$, let $\Delta = b^2 - 4ac$. If $\Delta < 0$, there is no real root; if $\Delta = 0$, there is one root $-\dfrac{b}{2a}$; if $\Delta > 0$, there are two roots $\dfrac{-b - \sqrt{\Delta}}{2a}$ and $\dfrac{-b + \sqrt{\Delta}}{2a}$. To order the roots, take the sign of a into account.

Status and justification. Multiplying the equation by $4a \neq 0$ gives $(2ax + b)^2 = \Delta$. A square is positive or zero: the three cases follow. If $\Delta > 0$, $2ax + b = \pm\sqrt{\Delta}$. Splitting into cases avoids taking the square root of a negative number.

Example 2. $2x^2 + x - 3 = 0$: $\Delta = 1 + 24 = 25$; $x = \dfrac{-1 \pm 5}{4}$, that is 1 and $-\dfrac{3}{2}$. Check: $(x-1)(2x+3) = 2x^2 + x - 3$.

Watch out. The denominator is $2a$, not 2; the sign of b is kept with its brackets in b^2.

Application. Solve $3x^2 - 6x + 3 = 0$ and $x^2 + 2x + 5 = 0$.

2.3 Factorising without automatically using the discriminant

Key point · Definition and results. If two real roots exist (possibly equal, in which case they are counted twice), their sum is $-\frac{b}{a}$ and their product is $\frac{c}{a}$. An obvious root or a remarkable identity often gives a shorter factorisation. The quadratic polynomials vanishing at two distinct reals r and s are exactly $a(x-r)(x-s)$ with $a \neq 0$.

Status and justification. Expanding $a(x-r)(x-s) = ax^2 - a(r+s)x + ars$ and comparing coefficients gives the relations. Conversely, if r and s have this sum and this product, expanding recovers the polynomial.

Example 3. Let $P(x) = x^2 - 7x + 12$. The numbers 3 and 4 have sum 7 and product 12, hence $P(x) = (x-3)(x-4)$. A separate question: determine a function f with roots 3 and 4 and $f(0) = 24$. Write $f(x) = a(x-3)(x-4)$. Then $12a = 24$, so $a = 2$: f differs from P.

Watch out. The sum of the roots is $-\frac{b}{a}$; do not forget the minus sign. The roots do not determine a: an extra condition is needed, for example the value of f at one point.

Application. Determine the polynomial f of degree 2 such that $f(1) = f(4) = 0$ and $f(0) = -8$.

2.4 Sign, inequalities and optimisation

Key point · Definition and results. For two distinct roots $r < s$, the quadratic has the sign of a outside $[r, s]$ and the opposite sign between the roots. It is zero at the roots. With no real root, it keeps the sign of a. With a double root, it keeps the sign of a away from that root. Optimisation is done on the domain of the problem, which may limit access to the vertex.

Status and justification. The sign table of the factors of $a(x-r)(x-s)$ gives the result. The vertex form compares $f(x)$ with β through the sign of $a(x-\alpha)^2$.

Example 4. $-2(x-1)(x-5) \geq 0$ is equivalent to $x \in [1,5]$. For $A(x) = x(12-x)$ on $[0,12]$, $A(x) = 36 - (x-6)^2 \leq 36$, with equality at 6. A rectangle with perimeter 24 m therefore has a maximum area of 36 m^2 when it is a square.

Watch out. In a strict inequality, the roots are excluded. Check the geometric bounds before concluding.

Application. Solve $x^2 - 4x - 5 < 0$, then maximise $x(8-x)$ on $[0,3]$.

Notes